

Source F — Findings Report

USDA ERS County Typology Codes (Structural Resilience Baseline)

Macro Embeddings project — internal technical report

Prepared 2026-07-13 · Data as of 2026-07-13

1. Executive Summary

Source F pulls the 2025-edition USDA Economic Research Service County Typology Codes for all 3,144 US counties/county-equivalents — the "Structural Resilience Baseline" anchor pillar of E_macro. Coverage is complete: every crosswalk county has a row, at the cost of a straightforward join decision (§2). Unlike Sources A and C, this is a single static file, so there is no missing-article or missing-series failure mode to characterize — the interesting findings instead concern how the *categorical* content of the typology actually splits the country, and how well its structural/demographic labels track the short-term economic momentum Source C measures.

Three findings stand out:

- **Half the country has no dominant industry (§3.1).** The proposal's framing of Source F as sorting counties into "Farming-dependent, Mining-dependent, Manufacturing-dependent, Government-dependent, or Recreation-dependent" undersells how common "none of the above" is: 50.0% of counties (1,572 of 3,144) are Nonspecialized, more than double the next-largest category (Manufacturing, 22.5%).
- **The exact same Connecticut geography-transition gap Source C found, independently surfaced here (§3.2).** All 9 counties ERS leaves with a not-classified industry-dependence sentinel are Connecticut's new Planning Regions — the identical root cause the Source C report documents for Connecticut's missing GDP series. Two unrelated federal data providers have the same lag catching up to Connecticut's 2022 county abolition.
- **Demographic distress flags don't predict short-term economic velocity (§3.4).** Cross-validating Source F's demographic risk flags (population loss, persistent poverty, etc.) against Source C's velocity metrics gives a near-zero correlation in both directions ($r = 0.007$ vs. unemployment velocity, $r = -0.019$ vs. size-normalized GDP velocity). This is a real, useful negative result: Source F is capturing *structural* character, not short-run momentum, and downstream consumers of E_macro should not expect the two pillars to substitute for each other.

The proposal's characterization of Source F as a stable, low-maintenance "baseline anchor" holds up well operationally — no API key, no rate limiting, one clean static download — but the *content* is more concentrated toward "unspecialized" and more structurally distinct from Source C's momentum signal than the proposal's narrative implies.

2. Methodology

2.1 Ingestion Pipeline

Script: `ingest_source_f.py` — downloads the ERS County Typology Codes CSV (`ers.usda.gov/media/6174/...-2025-edition.csv`), which is published long-format — one row per (county, attribute) pair across 13 attributes: six non-exclusive "high concentration" economic flags, one mutually-exclusive `Industry_Dependence_2025` code (0–5), six non-exclusive demographic risk flags, and a metro/nonmetro indicator.

The pipeline pivots this to one row per county, one-hot encodes the dependence code into `industry_dependence_{none,farming,mining,manufacturing,government,recreation}`, and

converts the already-binary flags to nullable booleans — ERS's sentinel codes (99 = not classified, -1 = insufficient data) map to null rather than False, so "no signal" is never silently conflated with "confirmed absent."

The raw ERS file has 3,152 FIPS codes, 8 more than the crosswalk's 3,144. Those 8 are Connecticut's legacy counties (FIPS 09001–09015), which the crosswalk no longer carries — Connecticut abolished its counties as administrative units in 2022 in favor of 9 new Planning Regions, and the crosswalk (built from the 2024 Census Gazetteer) uses the new geography exclusively. An inner join on `fips_code` handles this correctly with no special-case code: the legacy rows simply don't match anything and are dropped, while the 9 new Planning Region rows join cleanly. Result: **3,144/3,144 crosswalk counties covered, zero unmatched.**

Output: `data/source_f_usda_typology.parquet`, 3,144 rows, 21 columns (`county_name`, `fips_code`, `metro_2023`, 6 binary economic "high concentration" flags, 6 binary demographic flags, 6 one-hot `industry_dependence_*` columns).

3. In-Depth Findings

3.1 Dataset & Setup

Metric	Value
Counties in output parquet	3,144 (full US coverage, zero unmatched)
Source file	USDA ERS County Typology Codes, 2025 edition (static CSV)
Raw source rows (FIPS codes)	3,152 (8 legacy CT counties dropped on join)
Columns	21 (metro flag, 6 economic flags, 6 demographic flags, 6 one-hot dependence)
Auth / rate limiting required	None — single clean static download

3.2 Industry Dependence: "None" Dominates

Dependence category	Count	%
None (nonspecialized)	1,572	50.0%
Manufacturing	706	22.5%
Farming	354	11.3%
Recreation	267	8.5%
Government	146	4.6%
Mining	90	2.9%
Not classified (sentinel)	9	0.3%

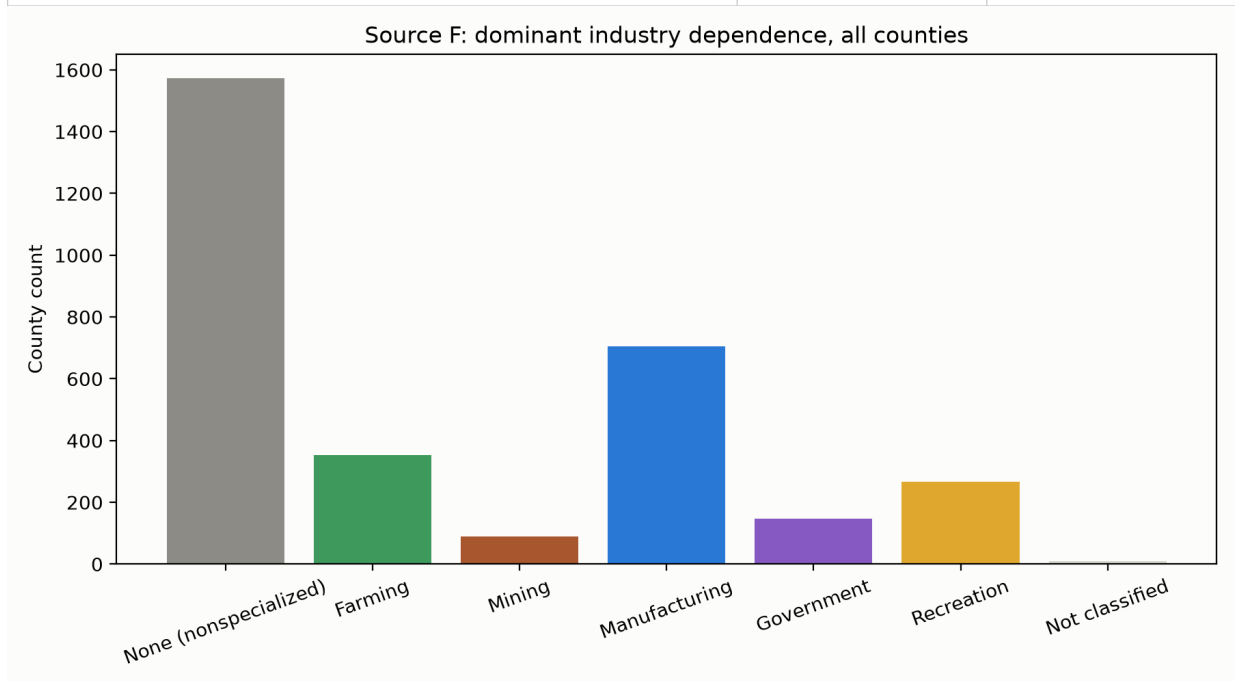


Figure 1. Bar chart of the 7-way industry dependence breakdown. Visually confirms that "None" is more than double the next-largest bar.

Half the country has no single industry earning/employment concentration high enough to be labeled dependent on it. This one-hot column is mutually exclusive by construction (verified: every county's six `industry_dependence_*` flags sum to exactly 0 or 1, never more), so this is a real distributional fact, not a data artifact.

The metro/nonmetro split explains part of it but not all: among *classified* counties, 67.6% of metro counties are "None" versus 39.6% of nonmetro counties — nonmetro counties are, as expected, more likely to have a concentrated economic base, but a plurality of them (39.6%) are still unspecialized even before counting metro counties. The proposal's framing ("groups counties into mutually exclusive economic dependencies") is accurate about the mechanism but implicitly undersells how large the residual "none of the above" category is.

3.3 The Connecticut Not-Classified Sentinel Matches Source C's Independent Finding

All 9 counties carrying ERS's 99 (not classified) sentinel for industry dependence are Connecticut's Planning Regions: Capitol, Greater Bridgeport, Lower Connecticut River Valley, Naugatuck Valley, Northeastern Connecticut, Northwest Hills, South Central Connecticut, Southeastern Connecticut, and Western Connecticut Planning Regions.

This is the same underlying cause the companion Source C report documents independently for its missing-GDP-series gap: Connecticut dissolved its traditional counties in 2022, and multiple federal statistical agencies — BEA/FRED for GDP and now ERS for typology classification — haven't yet backfilled data under the new Planning Region geography. Two unrelated sources landing on the same root cause is a useful cross-source confirmation that this is a genuine upstream data-lag issue, not a bug specific to either ingestion pipeline.

3.4 Nonmetro Counties Carry More Compounding Demographic Distress

Summing the six non-exclusive demographic flags (low postsecondary education, low employment, population loss, housing stress, retirement destination, persistent poverty) into a 0–6 "distress count" per county:

Distress count	Counties	%
0	1,194	38.0%
1	1,338	42.6%
2	392	12.5%
3	143	4.5%
4	60	1.9%
5	17	0.5%

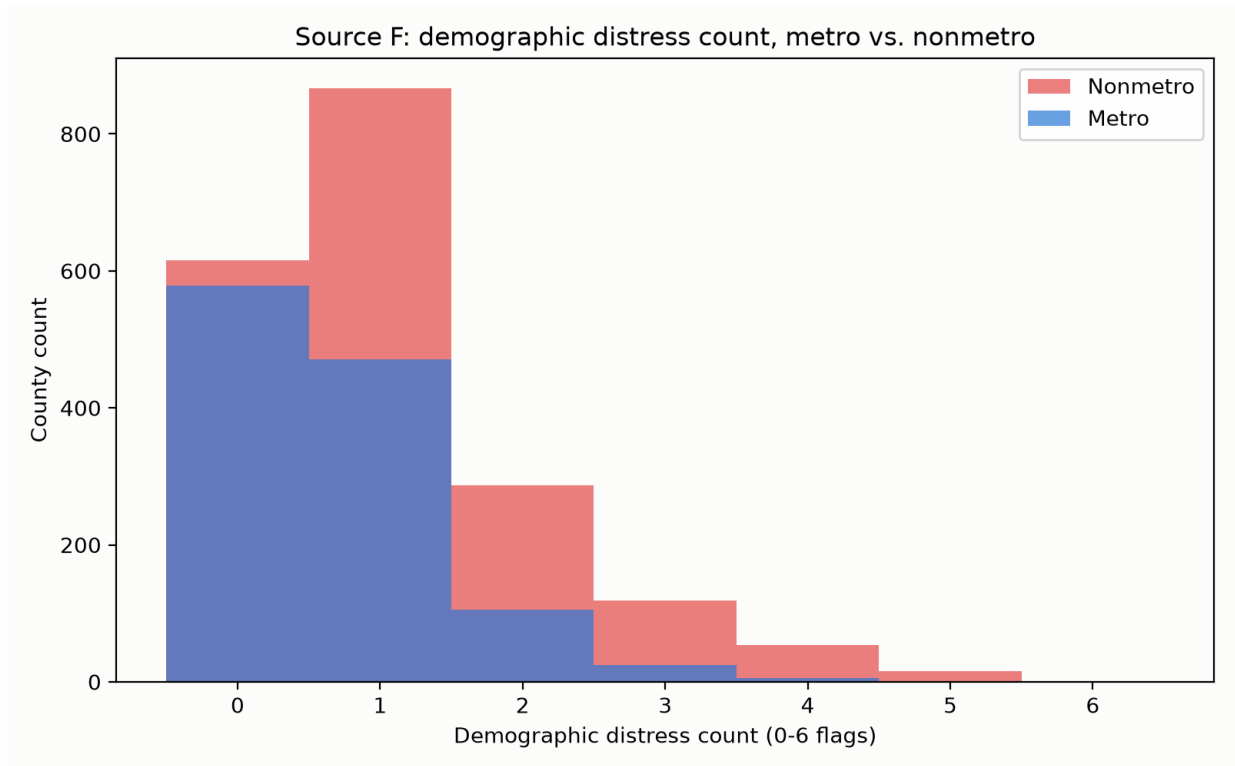


Figure 2. Stacked histogram of distress count, metro vs. nonmetro. Nonmetro's distribution is visibly right-shifted relative to metro's.

No county in this vintage carries all 6 flags — the observed maximum is 5. Mean distress count is 1.069 for nonmetro counties versus 0.659 for metro counties, a ~62% relative gap, in line with what the six flags are designed to capture (rural/nonmetro structural risk).

3.5 Cross-Validation Against Source C: Demographic Distress Does Not Track Short-Term Velocity

Joining Source F's distress count onto Source C's velocity metrics for all 3,144 counties (64 lack a `gdp_velocity` per the companion Source C report §2.2, so a size-normalized `gdp_velocity_pct = gdp_velocity / gdp_latest` — the same fix Source C's own findings recommended but hadn't yet implemented — is computed for this comparison rather than reusing the raw absolute-dollar column):

Comparison	Pearson r
Distress count vs. unemployment velocity	0.0065
Distress count vs. size-normalized GDP velocity	-0.0189

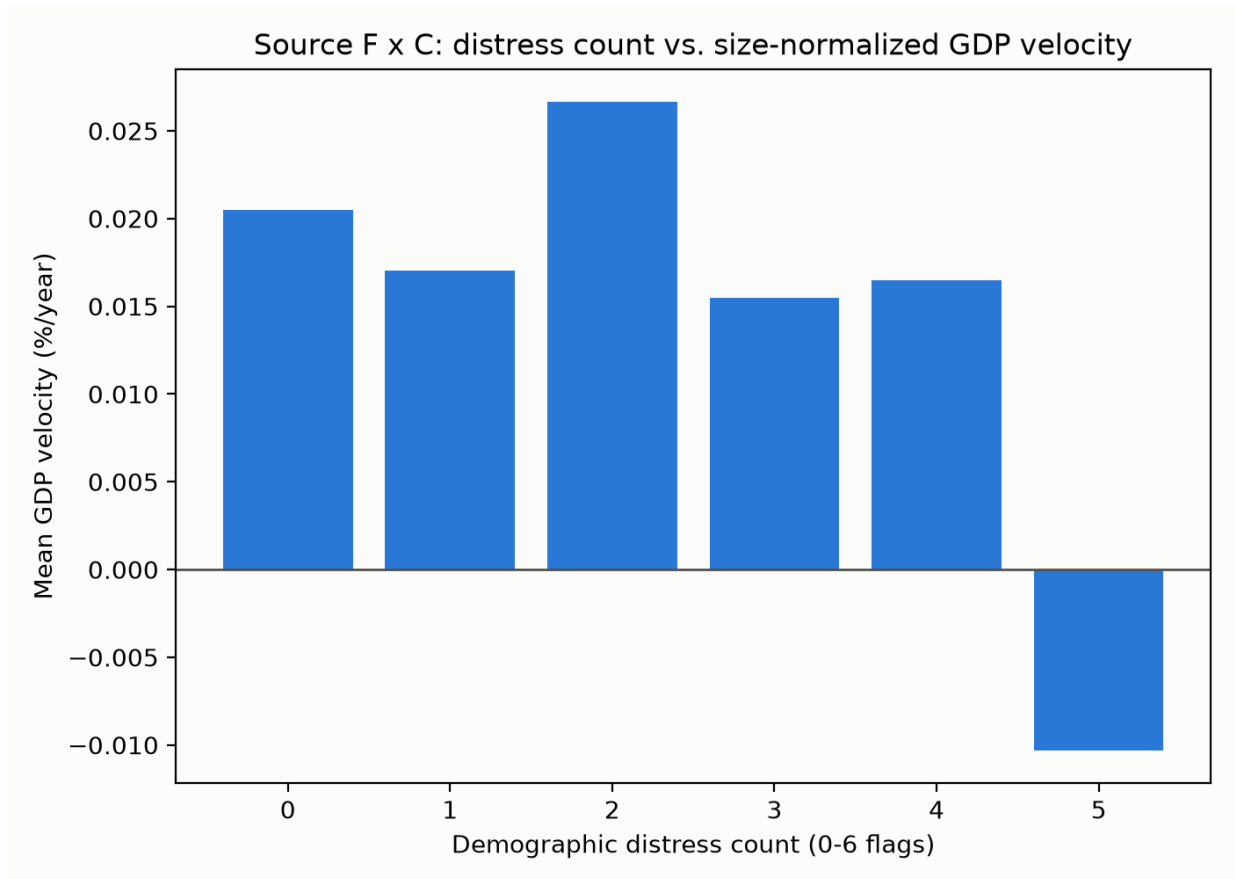


Figure 3. Mean size-normalized GDP velocity by distress count. Flat across 0–4, with a dip at 5 driven by a 17-county sample — the near-zero correlation visualized, including its noise floor.

Both correlations are indistinguishable from zero. This is a genuinely informative negative result: a county's structural demographic profile (population loss, persistent poverty, etc.) tells you essentially nothing about whether its economy is accelerating or decelerating *right now*. That's consistent with what the two sources are actually designed to measure — Source F is a slow-moving structural baseline, Source C is a 3-year momentum derivative — but it means E_macro consumers should treat them as genuinely complementary, orthogonal signals rather than expecting redundancy or reinforcement between them.

3.6 Limitations / Open Items

- Distress count treats missing flags as absent, not unknown — a small number of counties (8–33 depending on the flag, mostly persistent_poverty at 33) carry sentinel nulls, and the count is computed skipping nulls rather than penalizing or excluding, a likely slight undercount for the counties affected.
- The demographic flags are binary, not graded — "low employment" is a threshold crossing, not a magnitude, so the 0–6 distress count treats a county barely over a threshold the same as one far over it.
- Cross-validation (§3.5) is a single-round snapshot against Source C's current 3-year velocity window; the null result might not hold at longer horizons (e.g. 10-year population-loss trend vs. multi-decade GDP trend) — not tested this round.
- No validation yet against the proposal's specific "long-tail and rural markets" framing beyond the metro/nonmetro split in §3.2/§3.4 — a direct join against Source A (Wikipedia article length/existence) to test whether Source F really does anchor counties where Source A's text is thin would be a natural, low-cost follow-up.

4. Proposal Alignment Assessment

The `E_macro_extendedProposal.pdf` frames Source F as grouping counties into "mutually exclusive economic dependencies: Farming-dependent, Mining-dependent, Manufacturing-dependent, Government-dependent, or Recreation-dependent," positioned as a stable structural anchor especially valuable "for low-population or hyper-rural counties where online text data might be sparse or uninformative."

- **Supported:** the mechanism works exactly as described — the six dependence flags are genuinely mutually exclusive, and ingestion required none of the accommodations Sources A/C needed (no auth, no rate limiting, no per-county lookup table, no missing-series failure mode). The metro/nonmetro distress gap (§3.4) is consistent with the "rural resilience baseline" framing.
- **Complicated:** the proposal's five-category list reads as though most counties land in one of the five dependent buckets. In practice, half don't (§3.2) — "Nonspecialized" is the single largest category by a wide margin, which is worth surfacing explicitly for anyone using this as a categorical feature, since it means the modal county contributes no dependence-category signal at all.
- **New finding not anticipated by the proposal:** an independent, second data source (ERS) hitting the exact same Connecticut Planning Region data-lag gap Source C already found (§3.3) — a useful signal that this is a genuine upstream federal-data-lag issue worth flagging if `E_macro` ever needs full Connecticut coverage.
- **Not yet testable:** the proposal's stated Trade Logistics synergy with Source D ("establishes a baseline for how these regions absorb broader supply chain pressures") requires Source D, which doesn't exist yet. The Capital Flow synergy with Source A ("for... hyper-rural counties where online text data might be sparse") is directly testable now and flagged as next-round work.

5. Next Actions

- Once Source D (BTS FAF5) exists, test the proposal's stated Trade Logistics synergy directly.
- Check whether Source A's Wikipedia article length/intro quality is systematically thinner for the counties Source F flags as high-distress or non-metro — a direct test of the proposal's rural-baseline claim.
- No action planned this round on the Connecticut not-classified gap (§3.3) or the missing-flag undercount (§3.6) — revisit if a downstream consumer needs either fully resolved.

Appendix — Key Statistics

Metric	Value
Total counties	3,144
Not classified (industry dependence sentinel)	9
Dependence: None (nonspecialized)	1,572 (50.0%)
Dependence: Manufacturing	706 (22.5%)
Dependence: Farming	354 (11.3%)
Dependence: Recreation	267 (8.5%)
Dependence: Government	146 (4.6%)
Dependence: Mining	90 (2.9%)
"None" dependence rate, metro counties	67.6%
"None" dependence rate, nonmetro counties	39.6%
Mean distress count, metro	0.659
Mean distress count, nonmetro	1.069
Distress vs. unemployment velocity (Pearson r)	0.0065
Distress vs. size-normalized GDP velocity (Pearson r)	-0.0189

Source: [analysis-output/source_f_stats.json](#), [source-f-findings.md](#), [figures/source-f-numeric-summary.md](#) (MacroEmbeddings repo).